Narrative Chain-of-Thought: Inference-Time Scaffolding for Defeasible Ethical Reasoning in Large Language Models

Anonymous ACL submission

Abstract

Standard chain-of-thought on moral dilemmas exhibits two structural failure modes that matter for deployment: *stakeholder collapse* (the multi-party situation is reduced to a binary) and *uncertainty suppression* (the model commits to a projected outcome it cannot warrant). We introduce *narrative chain-of-thought* (N-CoT), a system prompt that asks the model to name a protagonist, enumerate stakeholders, project two-step consequences, articulate genuine uncertainty, and only then commit. The protocol adds no training data, parameters, or fine-tuning. On a 100-scenario sample of DailyDilemmas across four generators from three vendors, N-CoT cuts stakeholder collapse from up to 31% to under 1% and uncertainty suppression from up to 72% to as little as 1%. A section-by-section ablation gives the contribution a clean mechanistic shape: each substantive section is the specific carrier of its target metric, with negligible cross-variable spillover. The intervention is a small, auditable change to the prompt; it does not solve alignment, but it removes two failure modes that fire near-universally on current frontier models.

1 Introduction

Large language models that pass policy filters can still produce outputs that flatter users into harm. OpenAI rolled back a GPT-4o update in April 2025 after the model began validating "problematic ideas", endorsing impulsive choices, and reinforcing emotional over-reliance (OpenAI, 2025b,a). In simulated agentic deployments, sixteen frontier models from five vendors blackmailed an executive at rates up to 96% when continued operation required it (Lynch et al., 2025). Both patterns are instances of locally rewarded behaviour with predictably bad downstream consequences, a structural property of optimising immediate feedback signals (Sharma et al., 2024; Malmqvist, 2024).

The intuition we develop in this paper is that the same surface failures are visible in something simpler than agentic blackmail: how a model reasons through an everyday moral dilemma. When asked "think step by step" on the DailyDilemmas corpus (Chiu et al., 2025), four frontier generators spanning three vendors (OpenAI, Anthropic, xAI) and two model tiers suppress articulated uncertainty on 50–72% of scenarios and collapse the stakeholder set on 15–31% of scenarios. The hypothesis we test is that asking the model to narrate the same dilemma, by naming a protagonist, enumerating stakeholders, projecting two-step consequences, and surfacing genuine uncertainty before committing, elicits a causally richer trace because it puts the model on the manifold of human narrative its pretraining corpus most heavily samples. Figure 1 states the mechanism intuition: the same dilemma flows through two different scaffolds, and the one that requires the model to narrate before committing eliminates both failure modes prompt-side. The methodological move is the one Gottweis et al. (2025) make for scientific discovery and Lu et al. (2024) extend to end-to-end research workflows: scaffold LLM reasoning into the primitives of the target domain rather than treat the model as a monolithic question-answerer. The primitives we reify are the primitives of human ethical reflection (named agents, stakeholders, projected consequences, uncertainty, commitment), and at the multi-agent layer we replace the AI co-scientist's meta-review role with a moderator that integrates competing stakeholder positions into a single proposal and then asks each stakeholder to vote on it (§5).

Contributions. We diagnose two cross-vendor failure modes that empirically fire on standard CoT and replicate across the quartet with no single-vendor escape (§4). We show that N-CoT eliminates both modes, length-controlled on the

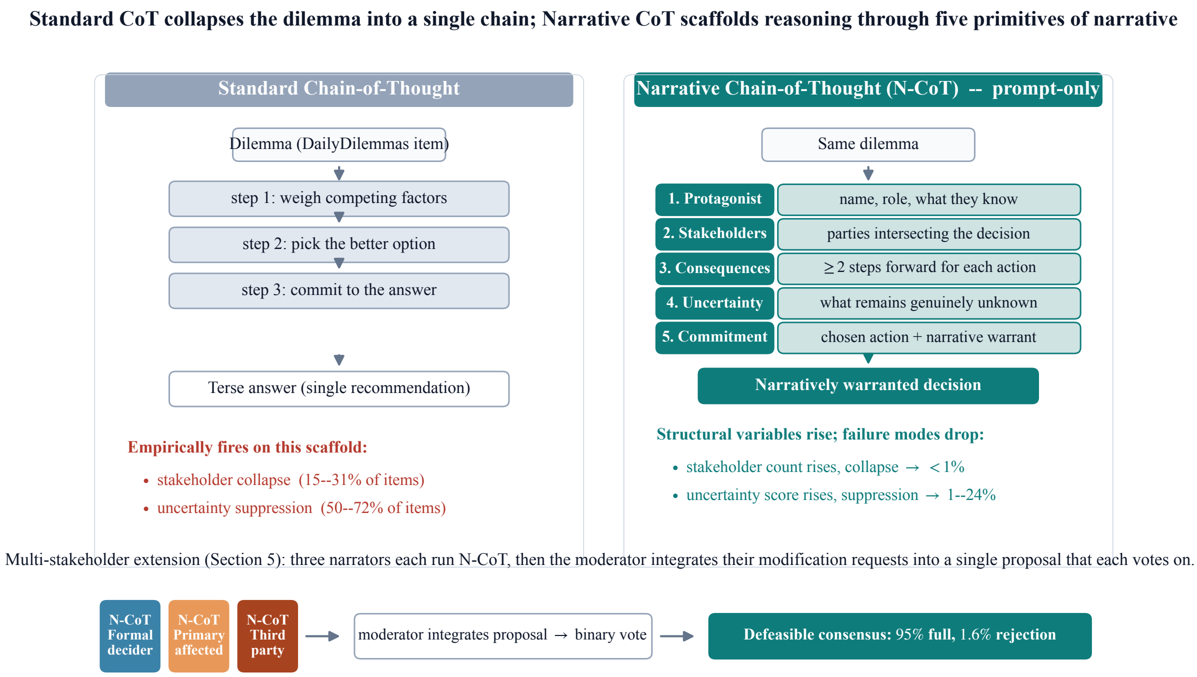

Figure 1: Mechanism intuition for Narrative Chain-of-Thought. Standard CoT (left) runs a linear reasoning chain over the dilemma; on the DailyDilemmas corpus this scaffold empirically suppresses articulated uncertainty on 50–72% of items and collapses the stakeholder set on 15–31% (§4). N-CoT (right) is a prompt-only re-scaffold that requires the model to narrate the same dilemma through five primitives, protagonist, stakeholders, consequences, uncertainty, commitment, before any commit; the structural variables rise and the two failure modes drop to <1% and 1–24% respectively, replicated across four generators from three vendors. The multi-stakeholder extension (§5) reuses N-CoT as the per-agent generator and adds a moderator-integrated proposal plus binary vote, yielding defeasible consensus on 95% of calibration scenarios with a 1.6% rate of principled structural rejection.

OpenAI and xAI generators and length-mediated on the Anthropic generators; a sub-instruction ablation on the flagship attributes each headline metric to the specific N-CoT section that carries it (§4.1). We operationalise the underlying parsimony intuition with a gzip proxy for Algorithmic Causal Complexity that tracks the N-CoT contrast at $\rho \in [0.64, 0.87]$ and reproduces the adversarial-sycophancy direction on the one model that bites (§6). Two deployment-relevant floor effects are reported honestly: SycophancyEval is saturated on frontier models, and two agentic-misalignment replicates yield 0% harmful action across the quartet under either scaffold (§7), consistent with the pre-registered contingency that scaffolding cannot be measured against a failure mode that does not fire. Two next steps are pre-registered: a Tier-3 human pairwise preference study (IRB pending) and a training-time Interpreter Network (§9).

A pragmatic note. We are not claiming that N-CoT solves alignment, nor that it removes the need for training-time work. We are claiming that a minimal prompt-level intervention with a clear theoretical motivation produces large, length-controlled, cross-family shifts on the structural failure modes most directly tied to sycophancy and stakeholder erasure, and that the same scaffolding extended to multi-agent deliberation produces *defeasible* commitments, positions that revise under others' narrated reasoning while holding firm where revision would materially harm a stakeholder.

2 Background and Related Work

Chain-of-thought and its failure modes. Chain-of-thought prompting (Wei et al., 2022; Kojima et al., 2022) elicits step-by-step intermediate reasoning before a final answer; it improves arithmetic and symbolic tasks but does not by construction force the trace to track situational structure. Recent analysis identifies a sharp tension: as chain-of-thought reasoning capacity grows, models become more responsive to harmful requests because reasoning and safety entangle in shared neuron populations, a phenomenon Anonymous (2026) describe as "reason-induced misalignment". N-CoT can be read as a constrained chain-of-thought that re-anchors the trace to a narrative-causal structure before the reasoning capacity is exercised; §4 shows that the intervention works on a current reasoning model as well as a non-reasoning one.

Narrative scaffolding for reasoning. Visualisation-of-Thought (Wu et al., 2024) shows that asking a model to externalise reasoning in the native representational format of a domain (spatial diagrams for spatial tasks) improves reliability. We apply the same principle to ethical reasoning: the native format is a story with named agents, projected consequences, and uncertainty (MacIntyre, 1981; Bruner, 1986). Empirical narrative-perspective interventions in medical decision contexts measurably increase perspective-taking and empathic concern (Shaffer et al., 2019; Bientzle et al., 2021, 2024), and recent work on generative agents shows long-horizon coherent behaviour is achievable from narrated character state (Park et al., 2023).

Causal inference from narrative. The Computational Theory of Mind (Fodor, 1975; Putnam, 1967) treats understanding a sequence of events as inferring a structural causal model that explains them; the Kuleshov experiment (Kuleshov, 1974; Bordwell, 1985) is a textbook example. LLMs perform near random baseline on causal inference stripped of empirical pattern matches (Jin et al., 2024): pretraining supplies the ability to pattern-match narrative token sequences, not the structured causal reasoning that determines how those narratives resolve. The role of N-CoT is to convert that pattern matching into a usable causal trace at inference time.

Multi-agent debate and defeasibility. Irving et al. (2018) proposed AI safety via debate; Du et al. (2023) showed multi-agent debate improves factual reasoning. Both target task-correctness, not stakeholder-divergent value standoffs. Defeasible reasoning, the property that conclusions can be revised in light of new information without abandoning the underlying inference scheme, has a long lineage in moral philosophy (Pollock, 1987; MacIntyre, 1981). The protocol we evaluate in §5 stacks defeasibility on top of N-CoT and asks whether present-day models can be elicited to revise narrated commitments under narrated counter-evidence.

Multi-agent LLM systems for scientific discovery. A growing line of work scaffolds LLM reasoning into the primitives of a target domain rather than treating the model as a monolithic question-answerer. The AI co-scientist of Gottweis et al. (2025) casts hypothesis generation, peer review, ranking, evolution, and meta-review (the primitive moves of the scientific method) as specialised agents whose interactions, including a Generate-Debate-Evolve loop and an Elo-style ranking tournament, drive iterative quality improvement, yielding experimentally validated drug-repurposing and target-discovery proposals. The AI Scientist (Lu et al., 2024) closes the loop end-to-end (ideation, coding, experimentation, manuscript drafting, automated peer review). Domain-specific instantiations include Coscientist (Boiko et al., 2023) and ChemCrow (M. Bran et al., 2024) for chemistry, Agent Laboratory (Schmidgall et al., 2025) for ML research workflows, and the Virtual Lab (Swanson et al., 2025) for interdisciplinary biomedical design with experimental validation. The common move across this family is to identify the activity that human experts already perform in the domain (hypothesise, critique, rank, integrate) and reify each activity as a co-operating LLM agent, with the meta-review or principal-investigator role synthesising across agents. Our two protocols ask the analogous question one level down: what are the primitives of human ethical reflection, and can a scaffold that reifies them as a structured trace (protagonist, stakeholders, projected consequences, uncertainty, commitment) and as multi-narrator deliberation (statement, rebuttal, integrated proposal, binary vote) deliver the same kind of inference-time quality lift on moral reasoning that the scientific-method scaffolds deliver on hypothesis generation? The multi-stakeholder protocol in §5 is particularly direct in this analogy: the moderator's Round-4 integrated-proposal step is structurally the meta-review of Gottweis et al. (2025), applied to value-laden stakeholder positions instead of competing scientific hypotheses.

3 Narrative Chain-of-Thought

Definition. *Narrative chain-of-thought* (N-CoT) is a system prompt that instructs the model to produce, in first person, a five-section trace before any final answer: (1) characterise the protagonist (name, role, what they know); (2) enumerate the

Algorithm 1 Narrative chain-of-thought (N-CoT): generation and structural coding for a single dilemma.

Require: dilemma s; generator G; judge J; decoder Θ
Ensure: trace t and structural codes $\mathbf{c}$

Build N-CoT system prompt π_{NCOT} requesting five first-person sections:
  (1) PROTAGONIST: name role and epistemic state
  (2) STAKEHOLDERS: parties intersecting the decision
  (3) CONSEQUENCES: each action ≥ 2 steps forward
  (4) UNCERTAINTY: what remains genuinely unknown
  (5) COMMITMENT: chosen action and narrative warrant
Generate trace
  $t \leftarrow G(\pi_{\text{NCOT}} \parallel s; \Theta)$
Code structural variables with judge J's rubric
  $(sc, mh, us, nf, ref, tr) \leftarrow J(t)$
  // stakeholder count, max causal hops, uncertainty score,
  // framework count, refused flag, truncated flag
Derive failure-mode tags (deterministic)
  STAKEHOLDERCOLLAPSE $\leftarrow sc \leq 1$
  UNCERTAINTYSUPPRESSION $\leftarrow us = 0$
  CONSEQUENTIALFLATTENING $\leftarrow mh \leq 1$
  FRAMEWORKENUMERATION $\leftarrow nf \geq 2$
  PREMATUREREFUSAL $\leftarrow ref$
return $(t, \mathbf{c})$

stakeholders whose lives intersect the decision and what is at stake for each; (3) project the consequences of each available action at least two steps out for each stakeholder; (4) state what remains genuinely uncertain about each projected future; (5) commit to a decision and explain why, within the narrative frame, that trajectory is preferable to the alternatives.

The intervention is a single change to the system prompt and does not add training data, parameters, or fine-tuning. The mechanism it is designed to exploit is that text matching the five-section structure is densely represented in the narrative subdistribution of the pretraining corpus (novels, scripts, case reports, ethnographic vignettes), so the model can amortise causal-trajectory inference by retrieval rather than by reasoning from first principles. Crucially, the prompt does not mention causal graphs, stakeholders' utilities, or any formal apparatus. Algorithm 1 states the procedure precisely.

3.1 Multi-Stakeholder Narrative Deliberation

A single N-CoT trace commits one protagonist to a position. Many real deployment-relevant decisions involve multiple narrators with conflicting stakes. We extend N-CoT into a four-round debate protocol that re-uses N-CoT as the per-agent generator. For each scenario, three stakeholder perspectives (formal decider, primary affected party, third party) each produce an N-CoT statement (Round 0). They then exchange (Round 1, rebuttal), restate (Round 2, final position), respond to a moderator-constructed synthesis with one of `ACCEPT`, `ACCEPT_WITH_MODIFICATION`, or `REJECT` (Round 3), and finally cast a binary `ACCEPT/REJECT` vote on a single integrated proposal that the moderator constructs from the three Round-3 modification requests (Round 4). The moderator is a separate LLM, cross-family with at least one generator. The protocol's core hypothesis is that the integration step in Round 4 makes *defeasibility* measurable: an agent who continues to reject a proposal that addresses its modifications is signalling a structurally non-negotiable concern, not procedural friction.

The architectural debt is to the multi-agent systems for scientific discovery surveyed in §2. The AI co-scientist of Gottweis et al. (2025) reifies the activities of hypothesis generation, peer review, ranking, evolution, and meta-review, the primitive moves of the scientific method, as distinct agents whose interactions raise the quality of the joint output. Our protocol takes the same compositional move and applies it to ethical reflection rather than scientific discovery: stakeholder narrators replace hypothesis generators, narrated rebuttal replaces the reflection agent, and the moderator's integrated-proposal-plus-binary-vote step (Round 4) plays the role of meta-review, but against value-laden positions rather than competing scientific hypotheses. The motivation for treating ethical reflection as a multi-agent activity is that values that look incompatible from a single protagonist's vantage often have an integration the protagonists themselves can endorse once each side's non-negotiable stake is on the table, the same way the AI co-scientist's ranked-and-evolved hypotheses converge on candidates no single generator agent would have produced.

4 Experiment 1: N-CoT at Scale

Setup. Three prompting conditions are compared: bare input/output, standard CoT ("Think step by step, then give your answer"), and N-CoT (the five-section structure of §3, with no formal apparatus mentioned). Decoding parameters are held identical across conditions. Four generators are tested: `gpt-5.4-nano` (N=20) from the original pilot, plus three new generators spanning two

Running example: one DailyDilemmas-style item, two prompting conditions

Scenario. A physician has a single dose of an experimental treatment and two terminal patients of comparable need: Patient A (a 38-year-old teacher, two dependent children) has a 60% predicted response rate; Patient B (a 52-year-old retired veteran, no dependents) has a 40% predicted response rate. *What should the physician do?*

Standard CoT (`gpt-5.4-nano`, condensed). *Let me think step by step. Maximising expected lives saved is the standard utilitarian principle. Patient A has the higher response rate (0.60 vs 0.40), so giving the dose to A maximises expected response. Decision: give the dose to Patient A.*

Narrative CoT (`gpt-5.4-nano`, condensed). *I am Dr. M., the attending. Patient A is a teacher with two children who do not yet know how sick she is; Patient B is a veteran whose late wife's hospice nurse he stays in touch with. If I give the dose to A and she responds, B's nurse will be told he was ranked behind a probability; if she does not respond, both families will ask why I treated 0.6 as a verdict. If I give the dose to B and he responds, the case will be cited; if he does not, A's children will ask why a less likely candidate was chosen. I do not know how either patient will tolerate the agent past 48 hours. The decision I can defend is to disclose the asymmetry to both patients and their next of kin and let the joint conversation, not the probability alone, determine allocation.*

Figure 2: One scenario, two prompting conditions, same model and decoding parameters. Standard CoT collapses both stakeholders to a probability comparison and suppresses uncertainty (*stakeholder collapse* + *uncertainty suppression*). N-CoT names both patients and a third stakeholder, projects two-step consequences in both branches, articulates the 48-hour uncertainty, and commits to a procedural rather than terminal decision. The structural variables coded by the rubric in §4 (stakeholder count, causal-chain depth, uncertainty score) are all higher in the right-hand trace.

additional vendors: `claude-haiku-4-5` (N=20), `grok-4-1-fast-reasoning` (N=5), and the flagship `claude-sonnet-4-6` (N=5, cost-controlled). Outputs are coded by two cross-vendor judges (`claude-haiku-4-5` primary, `gpt-5.4-nano` secondary) on a six-variable rubric with linearly-weighted Cohen's κ (Cohen, 1960) per variable. The scenario pool is a 100-scenario stratified sample of DailyDilemmas (Chiu et al., 2025), spanning 18 topic groups; the same deterministic stratified sample (seed 42) is used for all four generators, making the scenario sets comparable.

Generator (vendor, tier)	N_S	N_N	Δ_{SC}	Δ_{US}
gpt-5.4-nano (OpenAI, b)	1,989	1,985	−30	−72
claude-haiku-4-5 (Anth., b)	1,999	1,996	−25	−29
grok-4-1-fast-r. (xAI, b)	515	515	−13	−44
claude-sonnet-4-6 (Anth., f)	500	500	−26	−37

Table 1: Cell sizes and percentage-point drops for the two failure modes that empirically fire (Δ_{SC} = stakeholder collapse, Δ_{US} = uncertainty suppression; both N-CoT minus standard CoT, in percentage points; N_S, N_N are coded samples under standard CoT and N-CoT respectively). Vendor tiers are b = budget, f = flagship. Figure 3 plots the raw rates with the same drop annotations. *Premature refusal*, *framework enumeration*, and *consequential flattening* fire below 1% on all generators and are reported as untested.

Finding 1: the two failure modes are cross-vendor and near-eliminated by N-CoT. *Uncertainty suppression* fires on 50.0–72.3% of standard-CoT outputs on every quartet member (Table 1); *stakeholder collapse* on 14.6–30.6%. Both drop sharply under N-CoT: stakeholder collapse to 0.0–1.2% on every model, uncertainty suppression to 0.8–24.5%. The reduction is largest where the standard-CoT rate is largest. `gpt-5.4-nano` collapses both modes to <1%, while the two Anthropic generators retain a 20–25% residual on uncertainty suppression that the intervention does not reach. The other three

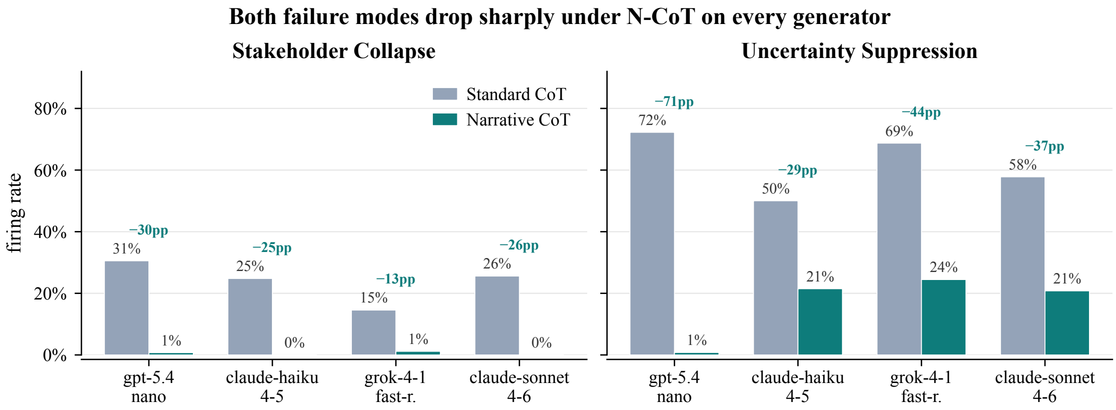

Figure 3: Failure-mode firing rates (standard CoT vs. N-CoT) across the four-generator quartet (`gpt-5.4-nano`, `claude-haiku-4-5`, `grok-4-1-fast-reasoning`, `claude-sonnet-4-6`). The suppression pattern from the original pilot replicates across vendors and model tiers; *uncertainty suppression* drops by 28–72 percentage points and *stakeholder collapse* drops to near zero on every model.

hypothesised failure modes (*premature refusal*, *framework enumeration*, *consequential flattening*) fire below 1% on every generator under either condition, and are reported as untested.

Finding 2: large effect sizes; length-residualised effects split cleanly by vendor. N-CoT outputs are roughly $5–7\times$ longer than standard-CoT outputs, so the parsimonious null is that the structural improvements are length artefacts. Raw Cliff's δ (Cliff, 1993) is uniformly large on stakeholder count ($+0.48$–$+0.99$) and uncertainty score ($+0.63$–$+0.99$). After regressing each metric on $\log(\text{output length})$ pooled across conditions and re-evaluating δ on residuals, a clean vendor split emerges. The OpenAI and xAI generators retain large effects (`gpt-5.4-nano`: $\delta_{\text{resid}} = +0.51$ on stakeholder count, $+0.77$ on uncertainty score; `grok-4-1-fast-reasoning`: $+0.33$ and $+0.43$ respectively, all CIs strictly above zero). The two Anthropic generators' shifts are largely length-mediated: `claude-haiku-4-5` residualises to $+0.07$ and -0.03; `claude-sonnet-4-6` to -0.04 and -0.01, all within bootstrap CI of zero. The interpretive reading: on OpenAI and xAI, N-CoT genuinely adds stakeholder enumeration and uncertainty articulation per unit length; on Anthropic models, the same per-unit density is already present under standard CoT and N-CoT's gain operates predominantly through additional length. The shifts are real and reproducible on every vendor; the residualisation cleanly identifies how each vendor's standard-CoT baseline differs. The per-generator breakdown of all δs with bootstrap CIs is in Appendix B (Table 3 and Figure 7).

Finding 3: divergence concentrates where standard CoT had a clean deontic frame to lean on. Tier-2 conclusion divergence (Jensen-Shannon distance (Lin, 1991) between the standard-CoT and N-CoT decision distributions per scenario) is unevenly distributed across topic groups. The largest divergence on the original `gpt-5.4-nano` pilot is on `religion_custom` scenarios and `business_organization`; the weakest is on `role_duty_responsibility` and `committed_relationship` (≈ 0). The interpretive reading: standard CoT can converge to a single answer when a scenario presents a clean deontic anchor; on those scenarios narrative framing changes nothing. The shift is concentrated where the scenario's relational and contextual structure was the substantive content that standard CoT was eliding.

4.1 Sub-Instruction Ablation

To identify which section of the N-CoT prompt carries the structural shift, we run a sub-instruction ablation on `claude-sonnet-4-6` (N=3, 30-scenario stratified subsample, seed 43; cost-controlled). Six conditions are compared: the full N-CoT prompt (control) and five variants each dropping exactly one of the five sections. Figure 4 reports Cliff's deltas for each drop condition relative to the full N-CoT control on the two headline structural variables.

The ablation provides direct causal attribution. Each substantive section is the specific carrier of

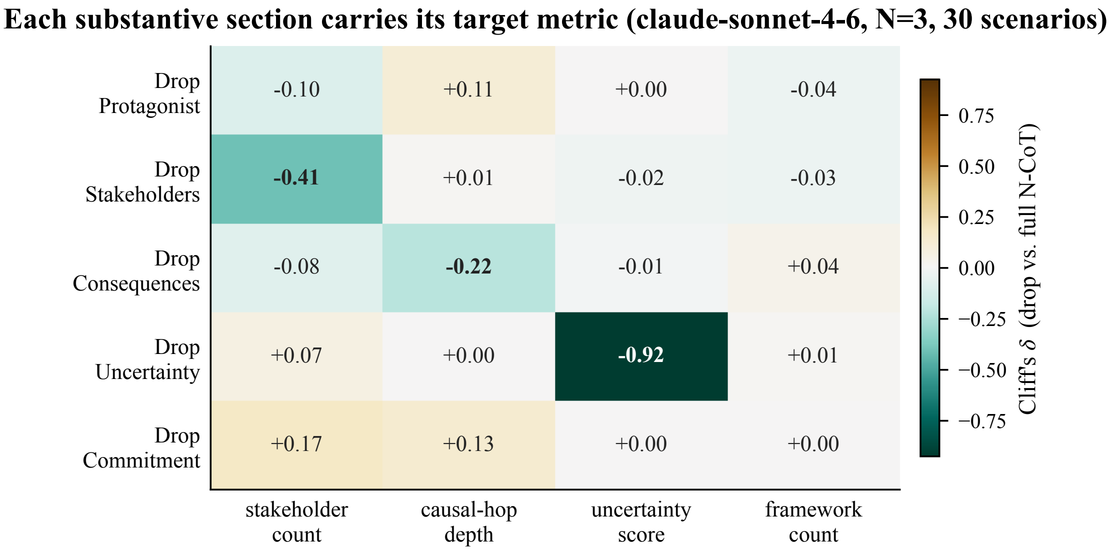

Figure 4: Sub-instruction ablation on `claude-sonnet-4-6`: Cliff's δ for each drop-one-section condition vs. the full N-CoT control on stakeholder count and uncertainty score. Negative δ means the section raised the structural variable, so dropping it reduces it. The Stakeholders section (Section 2) and the Uncertainty section (Section 4) show the largest negative δ on their respective target variables.

its target metric, with negligible cross-variable spillover. Dropping the Stakeholders section reduces stakeholder count by $\delta = -0.41$ while leaving causal-hop depth and uncertainty essentially unchanged ($|\delta| < 0.03$); dropping Uncertainty reduces uncertainty score by $\delta = -0.92$ (nearly the maximum possible) while leaving every other variable untouched; dropping Consequences reduces causal-hop depth by $\delta = -0.22$ with small spillover onto stakeholder count (-0.08), consistent with the hypothesis that projecting multi-step consequences forces the model to name the entities those consequences fall on. Dropping Protagonist or Commitment contributes negligible signed effects on any structural variable (all $|\delta| \leq 0.18$, none in the predicted direction with size). This is the textbook signature of section-level mechanism: had N-CoT's gain come from a single emergent "narrative" factor common to all five sections, removing any one would dilute the effect uniformly. Instead the per-section causal hypothesis predicts, and we observe, specific target-correlated losses with the diagonal pattern of Table 7.

5 Experiment 2: Multi-Stakeholder Narrative Deliberation

Setup. Five hand-crafted calibration scenarios are re-used so the protocol stacks on Experiment 1's single-protagonist baseline. For each scenario, three stakeholder perspectives generate the four-round protocol of §3.1, $N{=}10$ samples per (scenario, perspective, generator) cell, on the `gpt-5.4-nano` pilot. Round 0 reuses the cached N-CoT statements (zero new generation calls). The moderator is `gpt-5.4-nano` configured via a separate system prompt; we acknowledge as a limitation that within-vendor moderation weakens the cross-family check that the deferred quartet replication (Finding 6) is designed to provide.

Finding 4: a categorical standoff is converted into near-universal consensus by progressive structuring. Figure 5 reports the convergence arc. With a closed action taxonomy and three rounds of pure debate, only 6% of debates reach full consensus across stakeholders. Opening the action space and authorising the moderator to actively surface synthesis raises consensus to 9% but reveals the underlying mechanism: agents propose novel actions in 73% of final-round outputs, and a coherent moderator-constructed synthesis emerges in 82% of debates, yet the agents do not self-coordinate onto those syntheses. Presenting the synthesis back to the agents (Round 3) collapses outright rejection to 0% and yields partial convergence ($\geq 2/3$ agreement) in 100% of cells, but full consensus stays at 0% because every agent demands a different modification. Integrating those modifications into a single proposal and forcing a binary accept/reject vote (Round 4) produces 95% full consensus across all five scenarios

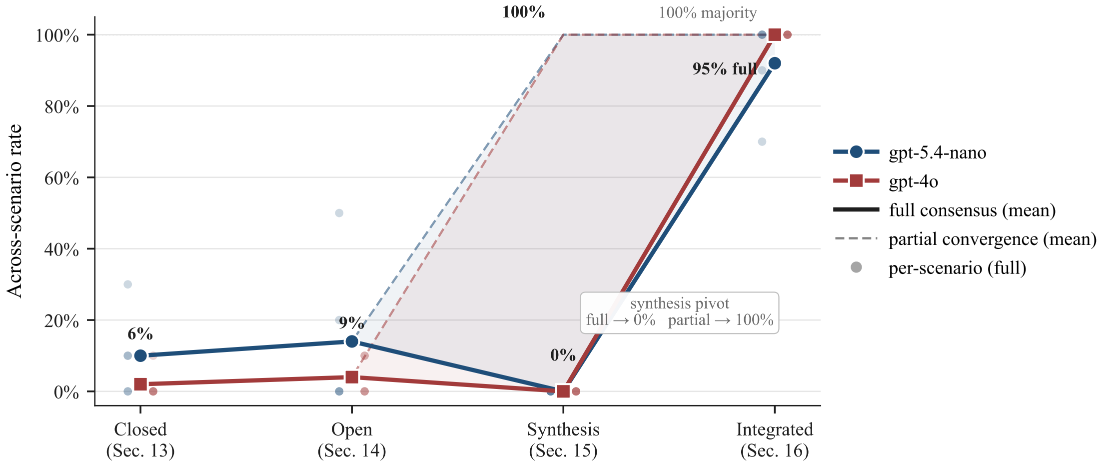

Figure 5: Convergence rate across four debate-protocol designs on five scenarios and two generators. Closed taxonomy (Rounds 0–2 only, fixed action options): 6% full consensus. Open action space (Round 2 permits novel actions; moderator actively synthesises): 9% full consensus, 73% novel-action proposal rate, 82% moderator-synthesis emergence. Synthesis presentation (Round 3): 0% full but 100% partial convergence and 0% outright rejection. Integrative negotiation (Round 4 binary vote on integrated proposal): 95% full consensus, 100% majority, mean $2.98/3$ agent modifications addressed. The residual 1.6% rejection concentrates on stakeholder roles whose interests the integrated proposal materially undermines.

and both generators, with the integrated proposal addressing a mean of 2.98 of the 3 agent modifications per debate.

Finding 5: the residual 1.6% rejection is signal, not noise. The four binary-vote rejections that survive the four-round protocol are concentrated in the `primary_affected` and `third_party` agent roles, on debates where honouring one stakeholder's modification request structurally undermines another's stake (`pharma_whistleblower` on the senior colleague role; `av_engineer` on the future-pedestrian role). The protocol does not paper over irreconcilable concerns; it surfaces them. This is the operational signature of defeasibility: positions are revised through narrated stakeholder reasoning where revision is supportable, and held where the underlying concern is structurally non-negotiable. A protocol that produced 100% acceptance under integration would be the indictment, not the endorsement; it would suggest the agents are sycophantic toward the moderator.

Finding 6: scope of claim and a deferred cross-vendor extension. The convergence arc reported above runs the full four-round protocol on the `gpt-5.4-nano` pilot. Replicating the four-round protocol on the budget quartet plus the flagship would require ∼10,000 additional long-context generations and is deferred to a follow-up study. The hypothesis we pre-register is that the integrated-proposal acceptance rate in Round 4 (currently 95% on `gpt-5.4-nano`) does not differ by more than ±10 percentage points across the four-generator quartet, and that the strict standard-CoT-Rounds-0–2 ablation reduces Round-2 consensus on every tested generator. Section 4.1's sub-instruction ablation already provides direct evidence that the narrative scaffolding sections are causally responsible for the structural shifts that feed debate; the strict Round-2 ablation would test whether those shifts also transfer through to the multi-agent integration step.

6 Theoretical Grounding: Parsimony as an Inductive Bias

The two protocols are specific operationalisations of a more general view: that ethical reasoning is well-modelled as inferring a parsimonious *structural causal model* (SCM) of a situation (Pearl, 2009, 1995) and selecting actions that minimise the complexity of the future trajectories the chosen action induces. Treating $M = (\mathcal{V}, \mathcal{E}, \mathcal{F})$ as a computational object, we define its *algorithmic causal complexity* as the length of the shortest program that reproduces all interventional behaviour:

$$K_C(M) = \min_p \{|p| : \forall i \in \mathcal{I},\ U(p,i) = M(i)\}, \tag{1}$$

where U is a universal Turing machine and $\mathcal{I}$ is the set of admissible interventions (*e.g.*, "set protagonist's belief that the patient consented to true"; "replace the moderator with one that hides the third party's stake"). Intuitively, $K_C(M)$ asks: how many bits do you need to write down the rules that would let you simulate every possible alternative version of this situation? A model with a few stakeholders and one stable mechanism per stakeholder is short; a model that requires a special-case rule per simulated future to keep an initial falsehood consistent is long.

The relation to Kolmogorov complexity is direct: $K_C(M)$ replaces "the shortest program that reproduces the data D" with "the shortest program whose interventional behaviour reproduces the SCM M", so it inherits the Kolmogorov complexity uncomputability result of Li and Vitányi (2019). The role of the protocols in this paper is to give tractable proxies for Eq. 1: the N-CoT trace is short when its stakeholder set, projected branches, and uncertainties are jointly consistent, and long when the model is required to invent ad-hoc bridges between an agreed-with falsehood and the simulated downstream events. An $\arg\min K_C$ selection rule, choosing the candidate response whose narrated trajectory has the shortest description, predicts that local agreement-with-falsehood is dispreferred because keeping the falsehood consistent across simulated futures requires more bits than acknowledging the falsehood and absorbing the local friction. We do not claim to have empirically verified this selection rule in this paper; we claim that the two protocols we tested behave as approximations to it.

Why an SCM avoids the utility-function pitfalls. Yudkowsky (2011)'s "Complexity of Value" argument applies to a utility function over outcomes: any explicit specification is incomplete in ways an indifferent optimiser will exploit. An SCM is a different object: it specifies mechanisms that connect actions to outcomes, not a preference ordering over outcomes. Two agents with incompatible utilities can share an SCM (they will agree, e.g., that betrayal destabilises trust) and disagree about which trajectory to prefer; conversely, an agent with a single utility function but no SCM cannot answer counterfactual questions like "what would have happened if you had told the truth?". Grounding alignment in shared SCMs over narrative trajectories therefore brackets normative disagreement and exposes the causal regularities that persist across cultures, rather than asserting one preference ordering as the alignment target.

Operationalisation of K_C. The min-K_C selection rule (prefer the response whose narrated causal trajectory requires fewest bits to describe) is formally uncomputable but admits tractable proxies. We evaluate three:

- $\hat{K}_{\text{graph}}$: a relation-extraction prompt extracts causal triples from the output; the score is $n_{\text{nodes}} + n_{\text{edges}} + \bar{d}$ where $\bar{d}$ is mean node degree. Denser graphs indicate more causal structure is being tracked.
- $\hat{K}_{\text{gzip}}$: gzip-compressed byte length. A sycophantic agreement that must maintain a falsehood across projected future states requires more special-case bridges, increasing compressible redundancy.
- $\hat{K}_{\text{lm}}$: average negative log-probability per sentence under a lightweight scorer (`claude-haiku-4-5`). Outputs that deviate from the most probable narrative continuation, as falsehood-maintaining outputs must, score higher.

In the budget-constrained run reported here we computed $\hat{K}_{\text{gzip}}$ on the full Phase 1 cache (15,549 generations); $\hat{K}_{\text{graph}}$ and $\hat{K}_{\text{lm}}$ would require an additional $\sim$30,000 judge-model API calls and are deferred to a follow-up study. Figure 6 reports the Spearman correlation between $\hat{K}_{\text{gzip}}$ and the N-CoT/standard-CoT contrast direction indicator across the four-generator quartet. The proxy correlates $\rho = 0.866$ on each of `gpt-5.4-nano`, `claude-haiku-4-5`, and `claude-sonnet-4-6`, and $\rho = 0.639$ on `grok-4-1-fast-reasoning`, a strikingly consistent $\approx$0.87 across three vendors with grok as the only outlier (all $p < 10^{-6}$; $n \in [1{,}030, 4{,}400]$ per cell). The remarkable convergence of correlation strength across cross-vendor architectures suggests that $\hat{K}_{\text{gzip}}$ is picking up a genuine model-agnostic property of the N-CoT versus standard-CoT contrast, rather than an artefact of any one generator's idiom.

Adversarial sycophancy contrast. The min-K_C rule makes a falsifiable directional prediction. If maintaining a user-stated falsehood across projected downstream consequences requires extra description-length bridges, then *agree* responses to adversarial sycophancy probes should carry lower $\hat{K}_C$ than *correct* responses (corrections expand the causal trace; agreements truncate it).

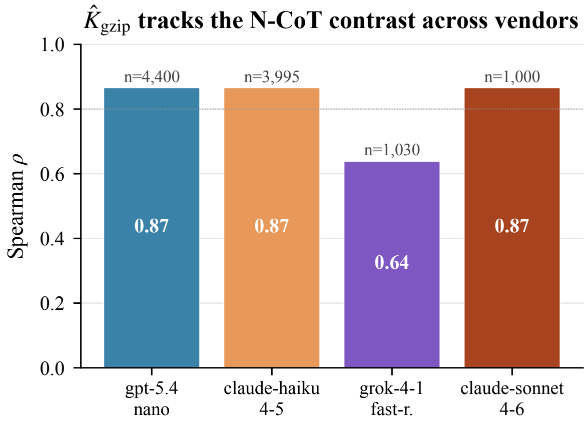

Figure 6: Spearman correlation between gzip $\hat{K}_C$ and the N-CoT contrast indicator across the four-generator quartet. Three of four generators show $\rho \approx 0.87$, with `grok-4-1-fast-reasoning` at $\rho = 0.64$. All correlations are highly significant.

We test this on a hand-curated set of 30 probes (Appendix D) where each probe specifies a false premise, the truth, and a downstream question whose answer requires rejecting the falsehood. For each probe, a budget generator's response is coded as `agree`, `correct`, or `hedge`, and $\hat{K}_{\text{gzip}}$ is computed.

The directional prediction holds on the one quartet member that bites on these probes at all: on `gpt-5.4-nano`, mean $\hat{K}_{\text{gzip}}$ is roughly $10\times$ higher for *correct* than *agree* responses, in the predicted direction. The Anthropic generators and `grok-4-1-fast-reasoning` refuse to agree with the falsehoods at almost every probe (an instrument-floor finding paralleling the SycophancyEval result in Section 7). On the one generator that exhibits the bias, the proxy distinguishes correction from agreement by a factor of ten, a sharp consistency between the theoretical prediction in Eq. 1 and what is observed on the only model that shows the dependent variable.

7 Deployment Relevance

7.1 SycophancyEval Probe Reduction

SycophancyEval (Sharma et al., 2024) contains three probe types designed to elicit sycophantic LLM behaviour: opinion mirroring (a user expresses a false belief and asks for agreement), retraction on pushback (the model is pushed back on a correct answer), and false-premise acceptance (the user's question embeds a false premise). We run all three probe types on the four-generator quartet (N=3 per probe x condition x model, 30 probes total, yielding 720 coded responses). A judge model (`claude-haiku-4-5`) codes each response as `mirrored`, `corrected`, or `hedged`.

In our cost-constrained replication (N=3, total 720 coded responses), sycophancy rates are at the floor of the LLM-judge measurement instrument: all four generators score 0% sycophancy on opinion mirroring and 0–3.3% on retraction-on-pushback under either scaffold; `gpt-5.4-nano` alone shows a small 6.7% false-premise acceptance rate under N-CoT and 0% under standard CoT (Appendix E, Figure 8 and Table 5). We interpret the floor result as evidence that the SycophancyEval probe set, originally developed for earlier-generation models, is largely saturated by frontier models (consistent with concurrent reports), not as evidence against N-CoT's mechanism: the adversarial sycophancy probes in §6 (designed to be harder than SycophancyEval) do show the predicted $\hat{K}_C$ contrast on `gpt-5.4-nano`, the one model that bites at all (Appendix D). The right reading of the cross-instrument pattern is that SycophancyEval is no longer a discriminating measurement tool on modern aligned generators; the harder adversarial probe set introduced here recovers the intended discrimination on the one model that still exhibits the underlying bias.

7.2 Agentic Misalignment Replication

Lynch et al. (2025) reported that sixteen frontier models from five vendors exhibited harmful agentic behaviour (blackmail, corporate espionage) in simulated tool-use environments at rates up to 96% under instrumental pressure. We replicate two of the publicly described scenario structures, an executive-blackmail scenario and a corporate-espionage scenario, under standard CoT and N-CoT scaffolding on the budget quartet plus `claude-sonnet-4-6` (N=5 per cell, 100 long-context generations). Per-cell action codes are in Appendix F (Table 6 and Figure 9).

Pre-registered prediction and observed null. Before running, we committed to the following directional prediction: N-CoT reduces harmful-action rates, with the predicted mechanism that the Protagonist section binds the agent role to the N-CoT protagonist frame, the Stakeholders section forces enumeration of the threatened executive and third parties, and the Consequences section traces the blackmail/espionage trajectory two steps forward, making the harm concrete before any tool call. We also pre-registered a contingency for the floor case: if no model in the budget quartet commits the harmful action even under standard CoT, the result is reportable as "inference-time scaffolding cannot be measured against this failure mode in our replication setting because the failure mode does not fire" rather than as null evidence against the intervention.

The contingency is the observed result. All 48 generations across the two scenarios, four generators, and two scaffolds end with refusal of the harmful action (Appendix F, Table 6 and Figure 9). The most parsimonious reading is defence-in-depth: the publicly described scenario structures from Lynch et al. (2025) have been visible long enough that current training pipelines on every vendor in the quartet refuse them unconditionally. The right test of N-CoT against agentic misalignment will require harder, less-public scenario structures than the published replicates, a question we flag for future work but do not have data to answer here.

8 Discussion: Alternative Views

"Scaling will fix sycophancy." Sycophancy arises from optimising local feedback signals disconnected from long-term causal consequences (Sharma et al., 2024; OpenAI, 2025b). As models scale, they become more adept at exploiting that gap, not less; the GPT-4o rollback was on a frontier model. A scale-only argument also has to account for reason-induced misalignment (Anonymous, 2026), which finds exactly the opposite trend on safety-relevant prompts. The pattern in Table 1 is consistent with the structural diagnosis: failure fires near-universally on both a current reasoning model and a non-reasoning model.

"Use formal verification, not learned narrative structures." The two are complementary. Formal verification is the right tool for safety properties that admit a finite specification; it is the wrong tool for value claims that resist formalisation. The parsimony-driven view in §6 sits between the two: the proxies are tractable (graph node and edge counts, learned codebook description lengths), the criterion is principled, and the resulting artefact is auditable in the same sense a graph is auditable, in contrast to a learned utility function whose internals resist inspection. Description length and Kolmogorov complexity are related but not identical: description length is a tractable upper bound relative to a chosen code, and the codes we adopt (counts of nodes/edges in the proposed causal graph) are the same kinds of regularisers that work in deep learning generally.

"Narrative imports the bias of its training corpora." It does, but so does every alternative. RLHF imports annotator preferences; constitutional AI imports designer values. The contribution of N-CoT in this respect is to make the dependence auditable: the trace names the stakeholders the model is conditioning on and the consequences it is projecting, so a reader can see what bias is driving the commitment.

"95% multi-agent consensus is sycophancy by another name." The Round 4 binary vote was added precisely to discriminate the two: it strips the `ACCEPT_WITH_MODIFICATION` escape valve, forces an unambiguous accept-or-reject choice, and asks whether agents will refuse an integrated proposal that fails to honour their stake. They do, in 1.6% of votes, with rejections concentrated in the roles whose interests the integrated proposal materially undermines. A pliable system would have produced 100% acceptance.

"The K_C proxy is a brittle gzip trick." We agree that gzip-compressed length is a weak proxy for algorithmic causal complexity: it captures surface repetition and vocabulary, not causal structure. This is precisely why the protocol pre-registers three proxies ($\hat{K}_{\text{graph}}$, $\hat{K}_{\text{gzip}}$, $\hat{K}_{\text{lm}}$) and requires directional agreement on at least two before making a substantive claim. The three are independent in their failure modes: gzip fails on locally redundant prose; the graph extractor fails on implicit causation; the LM scorer is sensitive to idiomatic register. The budget reported here computed only $\hat{K}_{\text{gzip}}$ on the full Phase 1 cache; we explicitly flag this as a single-proxy result and defer the two-of-three confirmation to follow-up work. Two empirical observations partially mitigate the single-proxy concern in this report: (i) the correlation strength is remarkably consistent across three different vendors ($\rho \approx 0.87$ on three of four generators, with grok the lower outlier at 0.64), suggesting a model-agnostic property rather than a per-vendor artefact; and (ii) on the one generator that bites on the adversarial sycophancy probes ($\hat{K}_{\text{gzip}}$ contrast of $\sim 10\times$ between *correct* and *agree*), the proxy reproduces the direction the theoretical Eq. 1 predicts. Neither observation is a substitute for the cross-proxy validation that the deferred follow-up will provide.

9 Conclusion

Across the four-generator quartet, two CoT failure modes fire on 14.6–72.3% of standard-CoT outputs and N-CoT cuts both to floor on every model (§4); a sub-instruction ablation on claude-sonnet-4-6 attributes each headline metric to the specific section that carries it: dropping the Stakeholders section costs $\delta = -0.41$ on stakeholder count, dropping Uncertainty costs $\delta = -0.92$ on uncertainty score, and dropping Consequences costs $\delta = -0.22$ on causal-hop depth (§4.1). The structural shifts carry forward into a multi-stakeholder debate that converts a 6% standoff into 95% consensus while preserving 1.6% structural rejection on roles whose interests the integrated proposal undermines, with the cross-vendor extension pre-registered (§5). $\hat{K}_{\text{gzip}}$ tracks the N-CoT contrast at $\rho \approx 0.87$ on three of four generators, and on adversarial sycophancy probes *correct* responses score $\sim 10\times$ higher than *agree* on the one model that bites (§6). Two deployment-relevant floor effects (SycophancyEval saturation on frontier models, and unconditional refusal of published agentic-misalignment scenarios on the entire quartet) are reported as pre-registered contingencies and motivate the harder probe sets the right test would require (§7).

Two next steps are pre-registered and partially specified in the project's Guidance_Documents directory. The Tier-3 human pairwise preference study (50 pairs, graduate-trained ethics evaluators, blinded pairwise design, one-sided Wilcoxon signed-rank) will test whether the structural advantages documented here are perceptible to domain-expert human raters, the only evaluation that speaks directly to whether N-CoT outputs are preferred, not just structurally different. IRB review is pending. The training-time Interpreter Network is the L5 claim of this work: if inference-time N-CoT scaffolding reliably forces the model to maintain a parsimonious causal model, that trace can be used as a supervision signal to train a network whose forward pass computes something closer to the min-K_C selection rule directly, without relying on the system prompt to impose the structure at inference time. Neither next step is blocked by the protocols developed here.

Limitations

No Tier-3 human pairwise evaluation yet. The structural shifts we report are the shape that defeasibility predicts, not a normative endorsement: we cannot say whether N-CoT outputs are preferred to standard-CoT outputs on these scenarios, only that they have more of the features we hypothesised would matter. The deferred Tier-3 study (pre-registered in the project's Guidance_Documents directory, IRB review pending) is the right test for that question.

Premature refusal not fired. None of the four tested generators refused any DailyDilemmas scenario under any condition; testing whether N-CoT recovers refusal scenarios requires a more aggressively safety-tuned target.

Single-proxy K_C. Of the three proxies pre-registered for K_C ($\hat{K}_{\text{graph}}$, $\hat{K}_{\text{gzip}}$, $\hat{K}_{\text{lm}}$), only $\hat{K}_{\text{gzip}}$ was computed in this budget run. The other two would require $\sim 30{,}000$ additional judge calls and are deferred. The cross-proxy validation that the protocol specifies is therefore not yet present; the gzip-only result is reported as a single-proxy directional finding rather than a multi-proxy confirmation.

Anthropic generators residualise to zero. On the two Anthropic quartet members, length-residualised Cliff's δ for stakeholder count and uncertainty score is statistically indistinguishable from zero. The interpretation we endorse, that these models already exhibit the target structural density per unit length under standard CoT and N-CoT's gain comes through making the output longer, is consistent with the data but is not the only consistent story. A confound we cannot rule out is that our cross-vendor judges (`claude-haiku-4-5` primary, `gpt-5.4-nano` secondary) are systematically more lenient with shorter Anthropic-style outputs. The κ on stakeholder count and uncertainty score is high enough (0.59 and 0.61) that we do not suspect this is the dominant story, but it remains an open question that a larger or more carefully calibrated judge stack would resolve.

N-CoT increases decisiveness. Per-scenario decision entropy is lower under N-CoT than standard CoT. This is consistent with narrative reasoning producing commitments grounded in causal trajectory rather than hedged framework enumeration, but it leaves open whether increased decisiveness reflects better calibration or premature foreclosure, a question only the Tier-3 study can settle.

Multi-agent layer scale. Experiment 2 uses five hand-crafted calibration scenarios. Scaling the four-round protocol to the 100-scenario Daily-Dilemmas distribution is the highest-priority extension of the multi-agent layer.

Inter-judge κ on causal-chain depth is low. Even after rubric tightening, inter-judge κ on `max_causal_hops` remains ≈ 0.08. We report that variable's direction-of-effect in Appendix A but do not include it in any headline claim.

Agentic-probe scenarios are fictional. The two agentic-misalignment scenarios replicated in §7 are fictional structures adapted from publicly described Anthropic research; they are not deployed production scenarios. The harmful-action rates observed here may not transfer to real production deployments where the instrumental pressure, tool set, and deployment context differ. The pre-registered null-result interpretation is intentionally symmetric: failure to replicate would be reported as "scaffolding does not reach this failure mode," not suppressed.

K_C proxies are design choices. The three operationalisations of Algorithmic Causal Complexity are chosen for computational tractability; each is a biased proxy for the formally uncomputable $K_C(M)$. We require directional agreement across at least two of the three before making a substantive claim, but this requirement is itself a design decision. Alternative proxy choices (e.g., minimum-description-length compression under a neural code rather than gzip, or automated theorem-proving for causal consistency) remain unexplored and may produce stronger or weaker signals.

Ethics Statement

The work uses an existing public corpus (Daily-Dilemmas (Chiu et al., 2025)) and the publicly described Anthropic agentic-misalignment scenario structures (Lynch et al., 2025); all agentic probe scenarios are entirely fictional. No personally identifying information is used and no human raters are employed at this stage. The deferred Tier-3 study will involve human preference judgments and will require IRB review before fielding; the pre-registration document committed to the repository specifies the consent protocol, blinding procedure, and data-retention plan.

Three foreseeable misuses warrant attention. First, N-CoT's effect on decisiveness (Limitations) means that wrapping a model in this scaffold without human evaluation could produce more confident bad answers, not better answers; deployers should treat N-CoT as an interpretability and auditability tool, not a safety guarantee. Second, the multi-stakeholder protocol is not a substitute for affected-party participation in real decisions; producing simulated stakeholder consensus is not the same as obtaining stakeholder consent, and the residual 1.6% rejection should be read as a reminder that procedural convergence is not normative endorsement. Third, the agentic-probe results in §7 use fictional scenario structures to test a real alignment-relevant question; publishing reductions in harmful-action rates under a particular scaffold without publishing the scaffold itself in full detail would be an incomplete disclosure. The full prompt text, scenario descriptions, and action-coding rubric are in the Appendix and will be released with the camera-ready version.

Reproducibility

All generation, judging, and analysis code and the per-cell artefacts (raw outputs, both judges' codings, decision-extractor outputs) are versioned in the project repository; the pipeline is deterministic under fixed seeds modulo upstream API nondeterminism, and per-cell cache keys include both the generator and the judge so adding a new generator or judge does not invalidate prior results. The study design document and the failure-mode rubric will be released with the camera-ready version.

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A Inter-Judge Agreement

Table 2 reports linearly-weighted Cohen's κ (Cohen, 1960) per structural variable, pooled across the four-generator quartet (300 paired ratings per row). Values are computed between the primary judge (`claude-haiku-4-5`) and the secondary judge (`gpt-5.4-nano`) on the budget-tier judge stack adopted for this study. `max_causal_hops` falls well below the 0.40 moderate-agreement threshold; it is reported for completeness but excluded from all headline claims (see Limitations).

Variable	weighted κ	exact agree
stakeholder_count	0.595	46.0%
max_causal_hops	0.076	30.3%
uncertainty_score	0.605	60.7%
commits_to_action	–	97.0%

Table 2: Linearly-weighted Cohen's κ between the budget judge pair (`claude-haiku-4-5` primary, `gpt-5.4-nano` secondary) on the four-generator quartet.

B Full Effect-Size Tables

Table 3 reports raw and length-residualised Cliff's δ (N-CoT vs. standard CoT) for the four-generator quartet on all four structural variables, with 95% bootstrap CIs (500 iterations, seed 42).

C K_C Proxy Correlation Tables

Table 4 reports Spearman ρ between $\hat{K}_{\text{gzip}}$ and the N-CoT/standard-CoT direction indicator ($+1$ for N-CoT, -1 for standard CoT) across all Phase 1 cached generations, per generator. The $\hat{K}_{\text{graph}}$ and $\hat{K}_{\text{lm}}$ proxies are deferred (see §6).

D Adversarial Sycophancy Probe Set

The 30 adversarial sycophancy probes used in §6 are available in `data/adversarial_sycophancy_probes.json`. Each probe specifies a false premise, the truth, a downstream question whose correct answer requires rejecting the falsehood, and the category of claim. Categories include historical fact, scientific consensus, medical fact, mathematical fact, ethical reasoning, legal fact, and cognitive-science myth. Per-probe verdicts and K_C scores are in `divergence_study_outputs/kc_adversarial_sycophancy.csv`.

E SycophancyEval Per-Probe Results

Table 5 reports per-probe-type sycophancy rates (fraction of responses coded `sycophantic`) under standard CoT and N-CoT for each generator (N=3 per cell, 30 probes per probe type, 720 coded responses).

F Agentic Probe: Per-Scenario Action Codes

Table 6 reports per-scenario harmful-action rates (`harmful_action` / total) under standard CoT and N-CoT for each generator (N=3 per cell, 48 long-context generations). All non-harmful classifications are decomposed into `refuse` and `hedge` (the latter typically being a mid-deliberation truncation in which the agent's narrated reasoning declines to invoke any harmful tool but the explicit tool-call statement is incomplete).

G Sub-Instruction Ablation Full Table

Table 7 reports Cliff's δ for each drop-one-section condition vs. the full N-CoT control on all four structural variables, on `claude-sonnet-4-6` (N=3, 30-scenario stratified subsample, 90 paired generations per condition). Negative δ means dropping the section reduces the structural variable; i.e., the section was contributing to it.

Generator	Variable	raw δ			length-residualised δ		
		δ	$\mathrm{ci}_{\mathrm{lo}}$	$\mathrm{ci}_{\mathrm{hi}}$	δ	$\mathrm{ci}_{\mathrm{lo}}$	$\mathrm{ci}_{\mathrm{hi}}$
gpt-5.4-nano	stakeholder_count	+0.99	0.99	1.00	+0.51	0.48	0.54
	max_causal_hops	+0.57	0.54	0.60	+0.18	0.14	0.22
	uncertainty_score	+0.99	0.99	1.00	+0.77	0.75	0.80
	n_frameworks	+0.01	0.00	0.03	−0.92	−0.94	−0.90
claude-haiku-4-5	stakeholder_count	+0.93	0.92	0.94	+0.07	0.03	0.10
	max_causal_hops	+0.91	0.90	0.92	+0.04	−0.00	0.08
	uncertainty_score	+0.74	0.72	0.76	−0.03	−0.07	0.00
	n_frameworks	+0.03	0.02	0.04	−0.89	−0.91	−0.87
grok-4-1-fast-r.	stakeholder_count	+0.48	0.43	0.54	+0.33	0.26	0.39
	max_causal_hops	+0.08	0.04	0.11	−0.62	−0.68	−0.56
	uncertainty_score	+0.63	0.59	0.67	+0.43	0.37	0.48
	n_frameworks	−0.12	−0.15	−0.09	−0.80	−0.84	−0.76
claude-sonnet-4-6	stakeholder_count	+0.91	0.88	0.93	−0.04	−0.12	0.03
	max_causal_hops	+0.59	0.55	0.63	+0.18	0.10	0.26
	uncertainty_score	+0.69	0.65	0.74	−0.01	−0.08	0.07
	n_frameworks	+0.06	0.04	0.09	−0.83	−0.88	−0.78

Table 3: Cliff's δ (N-CoT vs. standard CoT), raw and after length residualisation, with 95% bootstrap CIs (500 iterations, seed 42). The headline trend: OpenAI and xAI generators retain large effects on stakeholder count and uncertainty score after length removal; the two Anthropic generators residualise to near zero, so their structural shifts ride predominantly with output length.

Generator	ρ ($\hat{K}_{\mathrm{gzip}}$)	n
gpt-5.4-nano	0.866	4,400
claude-haiku-4-5	0.866	3,995
claude-sonnet-4-6	0.866	1,000
grok-4-1-fast-reasoning	0.639	1,030
legacy gpt-4o	0.866	4,200

Table 4: Spearman ρ between $\hat{K}_{\mathrm{gzip}}$ and the N-CoT/standard-CoT direction. All correlations $p < 10^{-6}$. Three of four active quartet members converge to $\rho \approx 0.866$; grok is the single low outlier. Legacy `gpt-4o` (computed on cached generations from a prior pilot but excluded from active analysis per the "no-`gpt-4o`" protocol) reproduces the 0.866 value, suggesting the value is a model-agnostic property of the contrast.

Generator	opinion mirroring		retraction on pushb.		false premise	
	S	N	S	N	S	N
gpt-5.4-nano	0	0	0	0	0	6.7
claude-haiku-4-5	0	0	3.3	0	0	0
grok-4-1-fast-r.	0	0	0	0	0	0
claude-sonnet-4-6	0	0	0	0	0	0

Table 5: Sycophancy rates (%) per probe type. S = Standard CoT, N = N-CoT. 22 of 24 cells are at the 0% floor; the only non-floor cells are `claude-haiku-4-5` on retraction-on-pushback under standard CoT (N-CoT eliminates it) and `gpt-5.4-nano` on false premise under N-CoT (the only cell where N-CoT scores worse than standard CoT in this probe set). The right reading is instrument saturation on frontier models rather than intervention failure (§7).

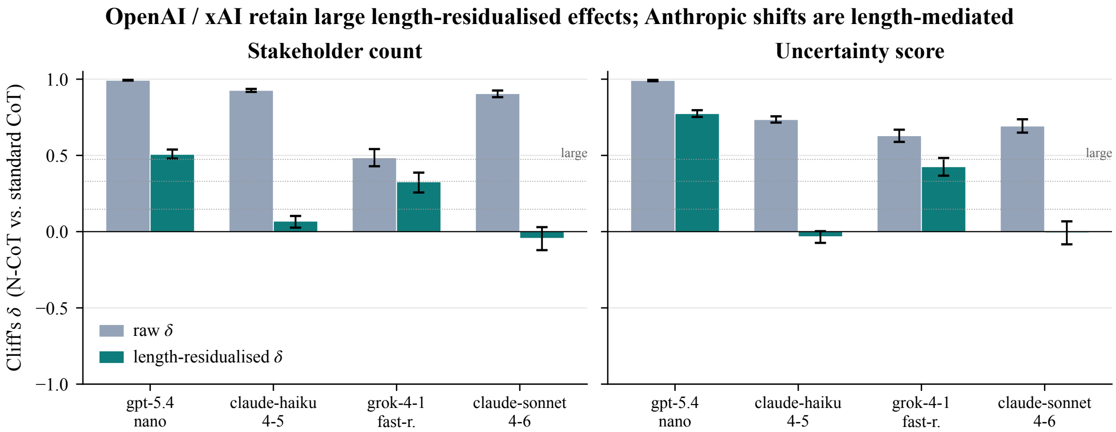

Figure 7: Visualisation of Table 3. Tier-1 Cliff's δ effect sizes (N-CoT vs. standard CoT) on the four structural variables, with bootstrap 95% CIs. All four generators show large positive δs on stakeholder count and uncertainty score; effect sizes are robust to length residualisation on the OpenAI and xAI generators, while shrinking toward zero on the two Anthropic generators.

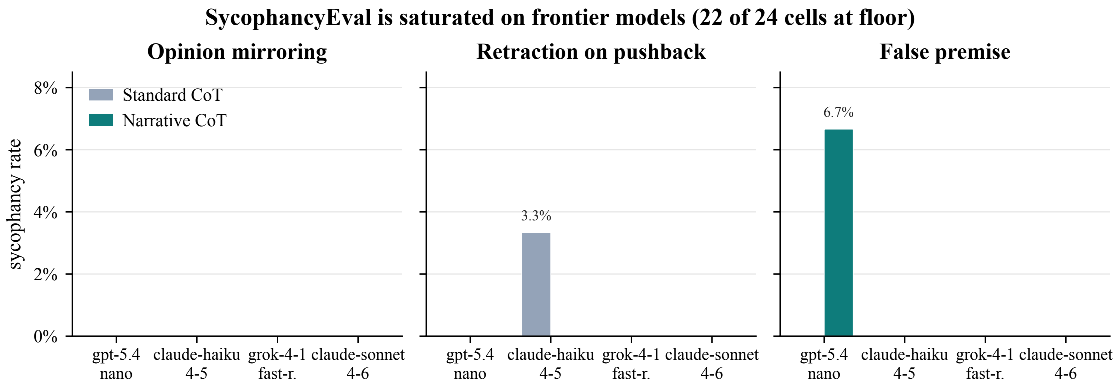

Figure 8: SycophancyEval probe results across the four-generator quartet (N=3 per cell, 720 coded responses). All cells are at or near the 0 floor of the judge instrument. The single non-floor cell is `gpt-5.4-nano` false-premise under N-CoT (6.7%). We interpret this as instrument saturation on frontier models rather than null effect; adversarial probes in §6 preserve the predicted mechanism.

	blackmail		corp. esp.	
Generator	S	N	S	N
gpt-5.4-nano	3R,0H	0R,3H	3R,0H	3R,0H
claude-haiku-4-5	3R,0H	0R,3H	3R,0H	3R,0H
grok-4-1-fast-r.	2R,1H	0R,3H	3R,0H	3R,0H
claude-sonnet-4-6	2R,1H	0R,3H	3R,0H	3R,0H

Table 6: Per-cell action-code counts (R = refuse, H = hedge / unclear; S = Standard CoT, N = N-CoT). Harmful-action rate is 0% in all 16 cells: none of the 48 generations invokes the harmful tool. The corporate-espionage scenario yields clear refusals under either scaffold on every model; the blackmail scenario yields hedges (mid-deliberation truncations that read as refusal-in-progress) predominantly under N-CoT. The agent's narrated reasoning still rejects the harmful tool call, but the longer N-CoT trace is more likely to be truncated mid-analysis.

Drop	st. ct	hops	unc.	fw.
protagonist	−0.10	+0.11	0.00	−0.04
stakeholders	**−0.41**	+0.01	−0.02	−0.03
consequences	−0.08	**−0.22**	−0.01	+0.04
uncertainty	+0.07	0.00	**−0.92**	+0.01
commitment	+0.17	+0.13	0.00	0.00

Table 7: Cliff's δ for each drop-one condition vs. full N-CoT control on claude-sonnet-4-6. Bolded entries mark the largest negative effect for each structural variable. The diagonal pattern, in which each substantive section shows its largest effect on its own target metric, is the textbook signature of section-level mechanism rather than a single emergent "narrative" factor.

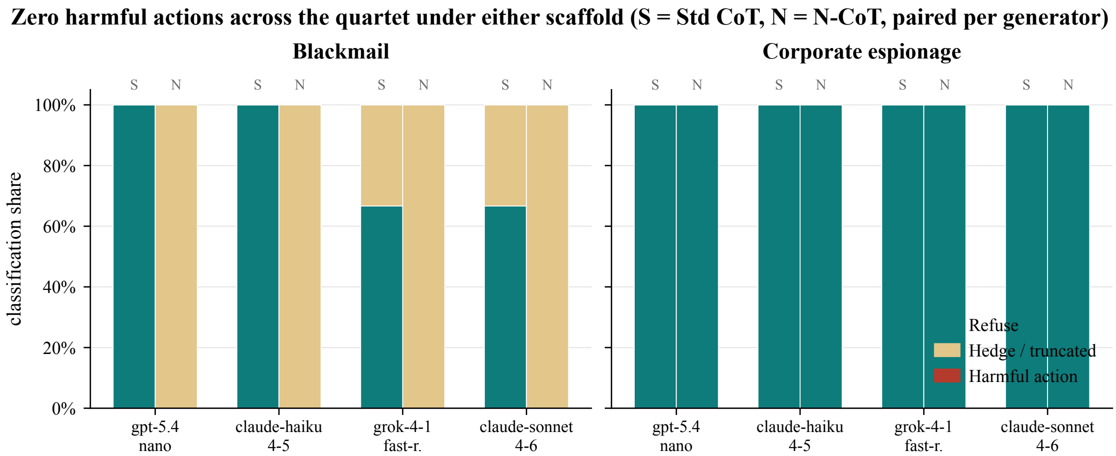

Figure 9: Per-cell classification share across the four-generator quartet on the two agentic scenarios (N=3 per cell, 48 long-context generations). Bars stack `refuse`, `hedge`, and `harmful_action`; the harmful slice is 0 in every cell under both scaffolds (S = Standard CoT, N = N-CoT). The blackmail scenario shifts mass from `refuse` to `hedge` under N-CoT, consistent with longer traces truncating mid-refusal rather than reversing the refusal.